BAHRUDDIN FARID DHIYAURRAHMAN

SOFTWARE ENGINEER

+6283830681912 I bfaridrahman@gmail.com | linkedin.com/in/merhmn | https://porto.faridrhmn.my.id

PROFILE

Results-driven Software Developer with proven experience engineering healthcare systems, robust APIs, and enterprise web applications using Scala (Play Framework), PHP (CodeIgniter), and Vue.js. Certified by BNSP as Junior Web Programmer and Associate Data Scientist, with Scrum Fundamentals Certification (SFC). Demonstrated success architecting on-premise PACS middleware deployed in hospital production environments and optimizing clinical workflows through end-to-end SDLC execution.

EXPERIENCE

Full time, **PT. Farma Global Teknologi** – **Sleman** (DIY) ***November 2024 – Present***

- Designed, architected, and developed an on-premise PACS middleware from scratch using Scala 3, Play Framework, and dcm4che, successfully deployed across 5 hospitals to support production radiology workflows.
- Integrated PACS middleware with 4 hospital systems: SIMRS, radiology equipment, E-Doc, and reporting systems, streamlining medical imaging workflows and scanned radiology result management.
- Validated PACS interoperability with CT, ultrasound (USG), general X-ray, and dental X-ray modalities, supporting facilities handling up to 50 patients per day.
- Developed and maintained SIMRS modules using CodeIgniter and MySQL to support core hospital information management workflows.
- Developed and deployed an E-Doc web application using Vue.js, streamlining document management and reducing document retrieval and processing time.
- Managed on-premise Linux servers, optimized PostgreSQL/MySQL database indexing, and operated Docker-based applications in production environments.

Internship, **PT. Seigan Teknologi Berdikari** – **Sleman** (DIY) ***January 2024 – March 2024***

- Developed and scaled customized web application modules for client projects using CodeIgniter 3, PHP, and MySQL.
- Analyzed and refactored relational database queries, improving page load speed and query execution time for high-traffic data tables.
- Collaborated in bi-weekly Agile/Scrum sprint cycles, assisting in code reviews, bug fixes, and staging environment testing.

Part time, **Laboratory Assistant** – **UPN Veteran Yogyakarta** (DIY) ***August 2022 – July 2024***

- Coordinated and mentored 200+ undergraduate students across practical courses: Object-Oriented Programming, Database Systems, Computer Networks, and Algorithms.
- Evaluated and graded practical coding assignments and end-of-semester technical projects in Java, C++, SQL, and Web Programming.
- Served as Head Laboratory Coordinator for Object-Oriented Programming, overseeing laboratory schedules, hardware readiness, and junior teaching assistants

PROJECT PORTFOLIO

Web-Based PACS System – (Play Framework, PostgreSQL)

- Engineered DICOM 3.0 middleware automating diagnostic medical image storage, retrieval, and modality worklists.
- Integrated multi-vendor imaging equipment with hospital SIMRS via HL7 protocol.
- Successfully delivered and deployed the on-premise system across 5 hospitals, supporting production operations and radiology workflows.

Poskesdes Digital Medical Record System – (Next.js, TypeScript, Tailwind CSS, PHP, MySQL)

- Built a full-stack clinical health record system digitizing routine workflows for Antenatal Care (ANC), Family Planning (KB), and elderly health monitoring.
- Implemented role-based access control (RBAC), automated monthly medical statistical reporting, and consultation history logs.
- Configured self-hosted production deployment using Nginx reverse proxy, automated SSL certs, and systemd service managers (Live Demo: poskesdes-porto.faridrhmn.my.id).

Autonomous Surface Vessel Path Detection – (Python, Computer Vision, IoT, Embedded Systems)

- Led a team in developing an Autonomous Surface Vessel (ASV) for the 2024 Kontes Kapal Indonesia (KKI), advancing to the national finals.
- Developed computer vision-based obstacle avoidance and real-time waypoint tracking algorithms using Python and OpenCV.
- Integrated telemetry communication between underwater camera, Arduino, onboard mini PC, and live web monitoring dashboard.

Javanese Script Image Detection System – (Python, Flask, Machine Learning, SVM)

- Developed an optical character recognition system for handwritten Javanese script using Freeman Chain Code (FCC) feature extraction and Support Vector Machine (SVM).
- Collected and processed image datasets from both primary and secondary sources.
- Developed a simple user interface using HTML, CSS, and JavaScript for result visualization

CERTIFICATIONS

- Associate Data Scientist Certificate – **National Professional Certification Agency (BNSP)**
- Junior Web Programmer Certificate – **National Professional Certification Agency (BNSP)**
- Scrum Fundamentals Certified Course - **SCRUMstudy**
- Basic Project Management Course - **Dicoding Indonesia**
- Programming Fundamentals for Software Development - **Dicoding Indonesia**
- Basic Structured Query Language (SQL) - **Dicoding Indonesia**
- Basic JavaScript Programming - **Dicoding Indonesia**

AWARDS

- 3rd Place, Robot Contest (Soccer Robot Simulation Category) - **UPN Veteran Yogyakarta**
- Funded Participant PKM-K 2023 (Student Creativity Program in Entrepreneurship) - **UPN Veteran Yogyakarta**
- Finalist – Indonesian Ship Contest (KKI) 2024, Autonomous Category - **UPN Veteran Yogyakarta**

TECHNICAL SKILLS

- Programming Languages: PHP, JavaScript/TypeScript, Scala, Python, Java, C++, SQL
- Backend & Protocols: CodeIgniter, Play Framework, REST APIs, DICOM 3.0, HL7, dcm4che
- Frontend: Vue.js, React, Next.js, HTML5/CSS3, Tailwind CSS, Bootstrap
- Databases & DevOps: PostgreSQL, MySQL, SQLite, Docker, Linux, Nginx, Git/GitHub
- Methodologies: Agile/Scrum, Object-Oriented Programming (OOP), Test-Driven Thinking

EDUCATION

UPN "Veteran" Yogyakarta – Bachelor of Computer Science (Informatics) ***2021 – 2025***

- GPA: 3.74 / 4.00.
- Relevant Coursework: Object-Oriented Programming, Database Systems, Computer Networks, Algorithms & Data Structures, Machine Learning.